UTSAV PIPERSANIYA

Data Engineer | Data Analyst — SQL, Python, ETL Pipelines, PostgreSQL
Jabalpur, Madhya Pradesh, India | +91 7477011067 | upipersaniya@gmail.com
LinkedIn: linkedin.com/in/utsavpipersaniya | GitHub: github.com/Utsav-x | Live: drops.raritaclub.com
Available immediately | Open to remote (India), or hybrid/on-site in Pune, Indore, Gurugram

SUMMARY

Data and backend engineer with nearly 3 years building **ETL pipelines, SQL data models and analytics services** in Python on PostgreSQL. Owns a fully automated extract–transform–load pipeline end to end at an enterprise ESG platform, plus the analysis engine that turns the loaded data into reported metrics. Strong SQL: cut query response times by ~50% through indexing, execution-plan work and ORM tuning. Experienced in data modelling, data cleaning and normalization, deduplication and entity resolution, idempotent incremental loads, and scheduled orchestration. Works in pandas and NumPy on time-series data, and translates reporting requirements from stakeholders into the schemas and pipelines that satisfy them.

TECHNICAL SKILLS

- **Data & SQL** — SQL, PostgreSQL, TimescaleDB, query optimization, indexing, execution plans, data modelling, schema design, star-schema thinking, window functions, aggregation
- **ETL & Pipelines** — ETL pipeline design, data pipelines, extract–transform–load, incremental loads, idempotent upserts, data cleaning and normalization, deduplication, entity resolution, fuzzy matching, data validation, data quality checks, scheduled orchestration (Celery Beat), pipeline monitoring and failure handling
- **Analysis** — pandas, NumPy, time-series analysis, Excel / openpyxl, reporting and export, metric definition, KPI aggregation
- **Languages** — Python, SQL, JavaScript, TypeScript
- **Backend & APIs** — Django, Django REST Framework, FastAPI, REST APIs, Celery, Redis, caching
- **AI for data** — Retrieval-augmented generation over pgvector, OpenAI API, natural-language querying of live platform data
- **Infrastructure** — Docker, Linux, Google Cloud Platform, Azure Blob Storage, Nginx, Gunicorn, Git, Sentry
- **Business** — Requirements gathering, stakeholder collaboration, reporting workflows, ESG and sustainability disclosure domain

EXPERIENCE

Backend / Data Engineer | Sustainext | Dec 2023 – Jul 2026

Enterprise ESG and sustainability reporting platform. Title is Backend Engineer; the work below is the data half of the role.

- **Owned a fully automated ETL pipeline end to end** — extracts data from an external third-party source, transforms it into the platform's schema, and loads it into PostgreSQL. Runs unattended on a schedule with no manual step, processing thousands of records per run. Making clients' third-party data onboardable **brought in 5 new clients**.
- Built an in-house **emission-factor reference database**, reducing reliance on — and cost of — a third-party emissions API.
- Built the pipeline's **transformation and data-quality layer in pandas and NumPy**: type coercion, normalization to the internal schema, validation of incoming records, and handling of malformed or partial source data so a bad upstream payload never corrupts the load.
- Built the **analysis engine** that turns loaded data into reported figures — automated emission calculation and aggregation across reporting periods, entities and categories, feeding enterprise ESG disclosures.
- **Optimized SQL performance: ~50% reduction in query and API response times**, measured before and after, through PostgreSQL indexing, elimination of N+1 access patterns, and ORM query tuning against real execution behaviour.
- **Cut batch processing time by ~60%** by restructuring long-running data jobs onto a Celery worker fleet with Redis, so large dataset processing stopped blocking user-facing reporting.
- Built **scheduled recurring jobs with Celery Beat** — the orchestration layer that runs ingestion, recalculation and reporting jobs on a cron schedule and surfaces failures.
- Designed the platform's **Redis caching layer**, including the invalidation strategy for figures that change as reporting data is revised.

- Built **Sustainext Chat**, an analytics assistant that answers questions about the customer's own ESG data. Retrieval-augmented generation over **pgvector** in PostgreSQL with **OpenAI** models, combined with live queries against platform data — effectively natural-language access to the reporting dataset for non-technical stakeholders.
- Delivered 100+ REST endpoints serving reporting and analytics workflows, working directly with frontend teams and translating reporting requirements into schemas, queries and endpoint contracts.

Freelance Backend Engineer | Rarita Technologies, Gurugram | Jul 2026 – Present

Collectibles marketplace and live-auction platform. Python, PostgreSQL, Celery Beat.

Reference Catalogue ETL & Entity Resolution

- Built a **CSV-to-database ETL importer** with a category-agnostic schema: core columns map to first-class fields, all remaining columns fold into a JSON attributes column, so new source formats load without a migration.
- Derived **normalized keys and a hashed deduplication key** as the natural key, so the bulk importer and the incremental fetcher agree on record identity and re-running the load is a no-op rather than a duplicate.
- Implemented **incremental, idempotent monthly upserts** from external sources on a Celery Beat schedule.
- Built **entity resolution with fuzzy matching**: exact-key lookup first, then a PostgreSQL `TrigramSimilarity` fallback bucketed by brand, scale and packaging, so variant spellings of the same item resolve to a single record.
- Added cleaning and coercion throughout — decimal parsing, year extraction, whitespace collapse, field-length clamping.

RaritaDrops — live at drops.raritaclub.com

- Shipped a live auction platform, including **post-sale settlement analytics** and a room-end summary with **Excel export** for operators.

PROJECTS

Index Options Time-Series Analysis

pandas, NumPy, Kite Connect API. Personal project.

- Analysed **minute-bar Nifty and Sensex data spanning 2018–2025** — roughly 1,800 trading days of intraday series — to develop and evaluate a strike-bucket options strategy.
- Ingested the instrument master via API, filtered and typed it, and **merged it against a constructed strike/expiry bucket frame** in pandas to resolve tradeable instruments.
- Computed indicator state per bucket with a cached carry-forward between runs so signals stayed consistent across restarts, and automated execution from the resulting signal.

EDUCATION

B.Tech — Gyan Ganga Institute of Technology and Sciences, 2023 | CGPA 8.34

CERTIFICATIONS

- **Data Analytics Virtual Internship** — Trainity, 2023 (Certificate ID 928726)
- **Technology Virtual Experience Program** — Deloitte via Forage, 2023 (Data Analysis, Coding, Development)